IE303 OPERATIONS RESEARCH I TERM PROJECT REPORT

Mini Logistics Decision-Support Tool for IUShipping

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1. Transportation Cost Estimation

Costs are estimated for a full round trip (going full, returning empty). Each trip's fixed cost (FC) combines fuel, driver wage, and tolls; dividing by truck capacity Q and adding a flat overhead O to give the per-ton Cij matrix.

$$FC_{ij} = D_{ij} \times (F_L + F_E) \times P_f + 2 \times T_{ij} \times W + 2 \times Toll_{ij}$$

$$C_{ij} = FC_{ij} / Q + O$$

Symbol	Parameter	Value / Unit	Source
Pf	Diesel price	3.20 BAM/L	GlobalPetrolPrices.com, Mar 2026
FL	Fuel cons. — loaded	0.32 L/km	Euro 6 truck benchmark
FE	Fuel cons. — empty (return)	0.24 L/km	≈ 25 % below loaded
W	Driver wage (employer)	15 BAM/h	Adorio.ba net × 1.65
Q	Truck capacity	24 tons	Standard 5-axle truck
O	Overhead per ton	5 BAM/ton	Maintenance, insurance, depreciation
M	Dummy-source penalty	9 999 BAM/ton	Large-M infeasibility guard

Distances from Google Maps; tolls from JP Autoceste FBiH Klasa IV schedule (Banja Luka routes estimated ~14.5 BAM via Autoput 9. januar).

- Unit-cost matrix Cij (BAM/ton, Normal – Pf = 3.20 BAM/L):

Origin\ Dest.	Tuzla	Zenica	Bihać	Brčko	Trebinje
Sarajevo	16.91	13.81	35.58	22.54	24.76
Mostar	28.89	23.64	33.50	34.38	16.10
Banja Luka	21.75	22.76	20.13	23.24	37.90

2. Transportation Model

Decision variables:

X_{ij} = tons of consumer goods shipped per week from production centre i to warehouse j

i = {Sarajevo, Mostar, Banja Luka} ; j = {Tuzla, Zenica, Bihać, Brčko, Trebinje}.

Objective Function: Min. $Z = \Sigma\Sigma\ C_{ij} \times X_{ij}$

Subject to:

Supply (each centre ships ≤ its capacity):	$\Sigma_j\, x_{ij} \le s_i$ for $i = 1, 2, 3$ (s = 250, 150, 200 t)
Demand (each warehouse receives exactly its demand):	$\Sigma_i\, x_{ij} = d_j$ for $j = 1\ldots5$ (d = 140, 120, 150, 100, 90 t)
Non-negativity:	$x_{ij} \ge 0$ for all i, j

Equality constraints are maintained by adding a dummy destination (cost 0) to absorb excess supply, and a dummy source (cost M = 9999) as an infeasibility guard for unmet demand. Sizes are computed automatically, so the manager can edit any supply/demand value without breaking feasibility.

Excel Solver configuration: Objective cell B20 (SUMPRODUCT of cost and decision matrices), Minimize; variable cells B12:G15 (x_{ij}); constraints as above; method: Simplex LP; Assume Non-Negative.

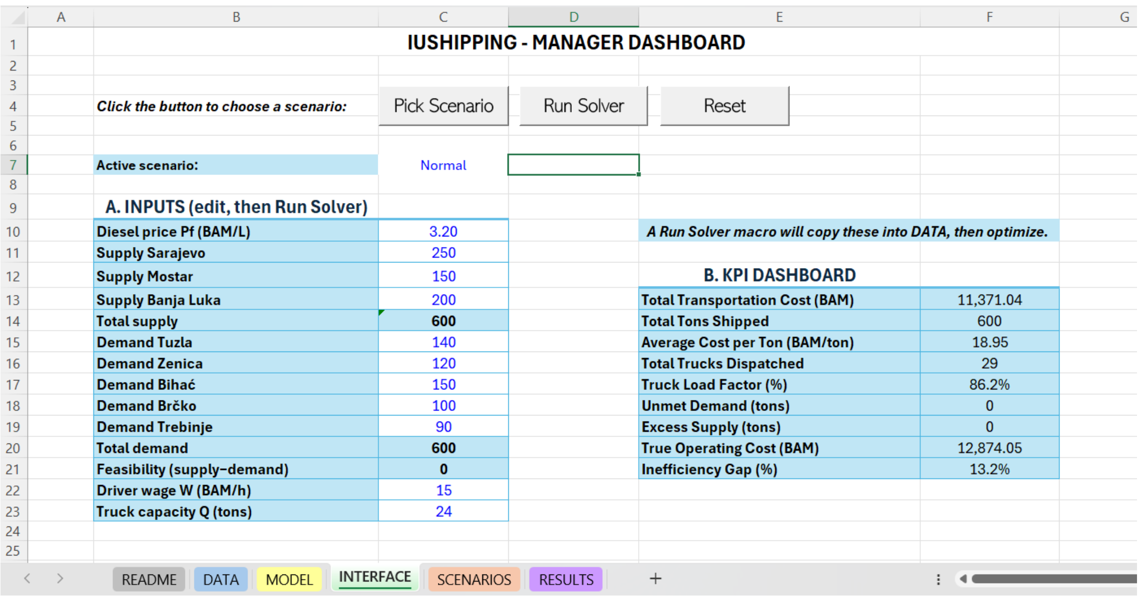
Optimal shipment plan – Normal baseline (Z^* = 11,371.04 BAM/week):

Origin\ Dest.	Tuzla	Zenica	Bihać	Brčko	Trebinje
Sarajevo	140	60	—	50	—
Mostar	—	60	—	—	90
Banja Luka	—	—	150	50	—

Trucks dispatched: 29 (from CEILING (x_{ij}/24)).

True integer-truck operating cost: 12,874.05 BAM – an inefficiency gap of 1,503.01 BAM (13.2 %) caused by partly-filled trucks on low-volume lanes. The LP optimum is reported as Z^*; the gap is documented separately in RESULTS sheet.

3. Screenshot of the INTERFACE:



4. Screenshot of the SCENARIOS:

C26 | The closer the trucks to be filled by 100%, the lower the avg. cost per ton is.

IUSHIPPING - SCENARIOS SHEET

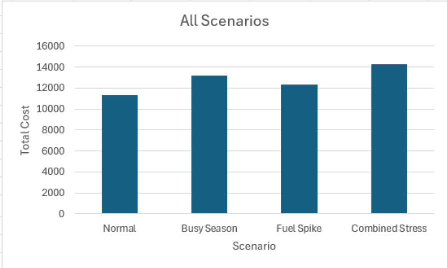
1. SCENARIO DEFINITIONS

Parameter	Normal	Busy Season	Fuel Spike	Combined Stress
Diesel price Pf (BAM/L)	3.20	3.20	3.68	3.68
Supply Sarajevo	250	315	250	315
Supply Mostar	150	190	150	190
Supply Banja Luka	200	250	200	250
Total supply	600	755	600	755
Demand Tuzla	140	182	140	182
Demand Zenica	120	144	120	144
Demand Bihać	150	180	150	180
Demand Brčko	100	100	100	100
Demand Trebinje	90	110	90	110
Total demand	600	716	600	716
Feasibility (supply−demand)	0	39	0	39

Busy Season raises supply too, so total supply ≥ demand and Z stays a real cost.*

2. SCENARIO RESULTS (run Solver once per scenario, then record here)

Metric	Normal	Busy Season	Fuel Spike	Combined Stress
Total cost Z* (BAM/week)	11371.04	13165.76	12319	14248.11
Cost per ton (BAM)	18.95	18.39	20.53	19.9
Trucks dispatched	29	33	29	39
Unmet demand (tons)	0	0	0	0
Notes	-	The closer the trucks	-	-



README | DATA | MODEL | INTERFACE | SCENARIOS | RESULTS

Conclusion and key observations:

Combined Stress scenario as the "compound" worst case (not just "combining both"), also to note that the busy season ships 11 % more tons yet achieves a lower cost per ton from 18.95 to 18.39 BAM because fuller trucks distribute fixed trip costs more efficiently.

Finally, the model achieves **Z* = 11,371.04 BAM/week** under normal conditions. Fuller trucks remain the strongest cost lever — the busy season lowers cost per ton despite higher total expenditure by improving load factors. The LP–integer gap (13.2 %; true cost **12,874.05 BAM**) is documented transparently in RESULTS as a known operational limitation.